How efficient is Fourier-Motzkin elimination as a computational tool? The main problem is that the number of inequalities generated by this method can go beyond all tractable limits within a few elimination steps. For this observe that if A has m rows, then $A^{/k}$ may have as many as $\lfloor \frac{m^2}{4} \rfloor$ rows: the number of inequalities can roughly be squared by every step, which leads to problems even with a fast implementation on a computer with a lot of memory.

Nevertheless the computations can be carried out fairly efficiently. See the notes to this lecture for comments about the available programs.

Let us mention only that Fourier-Motzkin elimination can in principle be used as an algorithm for linear programming (see Section 3.2). In fact, to find a point in $P(A, \boldsymbol{z})$ which maximizes $\boldsymbol{cx}$, we introduce an extra variable $x_0 = \boldsymbol{cx}$, and eliminate all the other variables. This will tell us the possible range of x_0, and by backtracking one can recover the optimal basis (i.e., a set of inequalities whose nonnegative combination yields an optimal upper bound on x_0). However, this method for linear programming is exponential, and there are much better ones available.

1.4 The Farkas Lemma

It was first pointed out by Kuhn [345] that with (termination of) Fourier-Motzkin elimination we have also done all the work for the *Farkas lemma*. This extremely important lemma appears in many different versions all over the theory of polytopes and polyhedra. It is interesting to note that if you look into different books and papers, you find quite different lemmas all called “the Farkas lemma.” All of these, however, are easily transformed into each other.

Essentially, the Farkas lemma yields a characterization for the solvability of a system of inequalities. There are variations for systems of inequalities in various standard forms: Farkas lemmas for polyhedra and for cones, for inequality systems with equalities, inequalities or strict inequalities, in nonnegative, positive or unrestricted variables, and so on. There are also quite different ways to formulate theorems “of Farkas type”:

- as *theorems of the alternative* (one inequality system has a solution if and only if a second system has none),
- as *transposition theorems* (because the second system can be derived by transposing the matrix and vectors of the first),
- as *duality theorems* (the duality theorem for linear programs is of Farkas type),
- as *good characterizations* (if a system has a solution, then any solution vector proves this; if it has no solution, then the Farkas lemma yields a dual vector that encodes this fact),

- as *certificates for validity* (if an inequality is valid for the solution set of a system, then it is a conical combination of the inequalities of the system),
- or, dual to this, as *separation theorems* (if a point does not lie in a convex-conical hull, then it can be separated from it by a linear functional).

We refer to Mangasarian [374] and to Stoer & Witzgall [529] for more versions, extensions, and generalizations. Separation theorems "of Farkas type" also hold for convex bodies. Infinite-dimensional versions are fundamental in functional analysis ("Hahn-Banach theorem").

Here is one basic version, characterizing the solvability of a general system of inequalities.

Proposition 1.7 (Farkas lemma I).
Let $A \in \mathbb{R}^{m\times d}$ *and* $\boldsymbol{z} \in \mathbb{R}^m$.
Either *there exists a point* $\boldsymbol{x} \in \mathbb{R}^d$ *with* $A\boldsymbol{x} \le \boldsymbol{z}$,
or *there exists a row vector* $\boldsymbol{c} \in (\mathbb{R}^m)^*$ *with* $\boldsymbol{c} \ge \mathbb{0}$, $\boldsymbol{c}A = \mathbb{0}$ *and* $\boldsymbol{c}\boldsymbol{z} < 0$,
but not both.

Proof. First observe that both conditions cannot hold at the same time: otherwise there are a column vector $\boldsymbol{x} \in \mathbb{R}^d$ and a row vector $\boldsymbol{c} \in (\mathbb{R}^m)^*$ with

$$0 \;=\; \mathbb{0}\boldsymbol{x} \;=\; (\boldsymbol{c}A)\boldsymbol{x} \;=\; \boldsymbol{c}(A\boldsymbol{x}) \;\le\; \boldsymbol{c}\boldsymbol{z} \;<\; 0,$$

which is a contradiction.

Now define $P := P(A, \boldsymbol{z})$, and $Q := P\big((-\boldsymbol{z}, A), \mathbf{0}\big)$. We note that an $\boldsymbol{x} \in \mathbb{R}^d$ exists with $A\boldsymbol{x} \le \boldsymbol{z}$ if and only if Q contains a point with $x_0 > 0$. Here Q is an $\mathcal{H}$-cone. Now we eliminate the variables $x_1, \dots, x_d$ from Q, to get the $\mathcal{H}$-cone $\mathrm{elim}_1\mathrm{elim}_2 \dots \mathrm{elim}_d(Q)$.

The key observation is that if we do Fourier-Motzkin elimination to get $\mathrm{elim}_i P(D, \mathbf{0}) = P(D^{/i}, \mathbf{0})$, then every inequality in the eliminated system $\mathrm{elim}_i(D)$ is a positive combination of at most two rows of D, so $D^{/i}$ can be written as $C^i D$ for a matrix C^i with only nonnegative entries, of which at most two per row are nonzero.

Iterating this idea, we get

$$\begin{aligned} \mathrm{elim}_1\mathrm{elim}_2 \dots \mathrm{elim}_d(Q) \;&=\; P((-\boldsymbol{z}, A)^{/d/d-1\dots/2/1}, \mathbf{0}) \\ =\; P(C^1C^2 \dots C^d(-\boldsymbol{z}, A), \mathbf{0}) \;&=:\; P(C(-\boldsymbol{z}, A), \mathbf{0}), \end{aligned}$$

where C is a nonnegative matrix.

All the inequalities in the system $C(-\boldsymbol{z}, A) \le \mathbf{0}$ are of the form $\gamma_{i0}x_0 \le 0$, since all variables other than x_0 have been eliminated.

Now assume that $P = \emptyset$, so that $Q \subseteq \{\boldsymbol{x} \in \mathbb{R}^{d+1} : x_0 \le 0\}$. By elimination we get

$$\mathrm{elim}_1\mathrm{elim}_2 \dots \mathrm{elim}_d(Q) \;\subseteq\; \{\boldsymbol{x} \in \mathbb{R}^{d+1} : x_0 \le 0\},$$

and thus the system $C(-\boldsymbol{z}, A)\boldsymbol{x} \leq \mathbf{0}$ contains an inequality $\gamma_{i0}x_0 \leq 0$ with $\gamma_{i0} > 0$. Let $\boldsymbol{c}$ be the row of C that yields this, then we have $\boldsymbol{c}(-\boldsymbol{z}, A) = (\gamma_{i0}, \mathbb{O})$, that is, $\boldsymbol{cz} = -\gamma_{i0} < 0$ and $\boldsymbol{c}A = \mathbb{O}$. □

Now comes another version of the Farkas lemma, for nonnegative solutions of systems of inequalities. Every such system can be rewritten as a system of inequalities, and this is exactly what we do to prove it. In fact, the proof nicely illustrates various simple-but-important techniques for rewriting systems of inequalities, such as the introduction of slack variables, and rewriting unbounded variables as differences of nonnegative ones.

Proposition 1.8 (Farkas lemma II).
Let $A \in \mathbb{R}^{m\times d}$ and $\boldsymbol{z} \in \mathbb{R}^m$.
Either *there exists a point $\boldsymbol{x} \in \mathbb{R}^d$ with $A\boldsymbol{x} = \boldsymbol{z}$, $\boldsymbol{x} \geq 0$,*
or *there exists a row vector $\boldsymbol{c} \in (\mathbb{R}^m)^*$ with $\boldsymbol{c}A \geq \mathbb{O}$ and $\boldsymbol{cz} < 0$,*
but not both.

Proof. We have the following equivalences: $\exists\, \boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{z},\ \boldsymbol{x} \geq \mathbf{0}$

$$\begin{aligned}
&\iff && \exists\, \boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{z},\ (-A)\boldsymbol{x} \leq -\boldsymbol{z},\ -\boldsymbol{x} \leq \mathbf{0} \\
&\iff && \exists\, \boldsymbol{x} : \begin{pmatrix} A \\ -A \\ -I_d \end{pmatrix} \boldsymbol{x} \leq \begin{pmatrix} \boldsymbol{z} \\ -\boldsymbol{z} \\ \mathbf{0} \end{pmatrix} \\
&\overset{\text{FL I}}{\iff} && \nexists\, \boldsymbol{c}_1 \geq \mathbb{O}, \boldsymbol{c}_2 \geq \mathbb{O}, \boldsymbol{b} \geq \mathbb{O} : \\
& && \qquad (\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{b}) \begin{pmatrix} A \\ -A \\ -I_d \end{pmatrix} = \mathbb{O},\quad (\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{b}) \begin{pmatrix} \boldsymbol{z} \\ -\boldsymbol{z} \\ \mathbf{0} \end{pmatrix} < 0 \\
&\iff && \nexists\, \boldsymbol{c}_1 \geq \mathbb{O}, \boldsymbol{c}_2 \geq \mathbb{O}, \boldsymbol{b} \geq \mathbb{O} : (\boldsymbol{c}_1 - \boldsymbol{c}_2)A - \boldsymbol{b} = \mathbb{O},\ (\boldsymbol{c}_1 - \boldsymbol{c}_2)\boldsymbol{z} < 0 \\
&\iff && \nexists\, \boldsymbol{c} = \boldsymbol{c}_1 - \boldsymbol{c}_2, \boldsymbol{b} \geq \mathbb{O} : \boldsymbol{c}A - \boldsymbol{b} = \mathbb{O},\ \boldsymbol{cz} < 0 \\
&\iff && \nexists\, \boldsymbol{c} : \boldsymbol{c}A \geq \mathbb{O},\ \boldsymbol{cz} < 0.
\end{aligned}$$

□

The following is my favorite version of the Farkas lemma. It says that if an inequality is valid for a polyhedron, then either it can be obtained as a positive combination of inequalities that define the polyhedron, or the polyhedron is empty, in which case the inequality $\mathbb{O}\boldsymbol{x} \leq -1$ can be obtained as a positive combination. This version of the Farkas lemma includes version I as a special case: $\mathbb{O}\boldsymbol{x} \leq -1$ is valid for all points $\boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{z}$ if and only if $A\boldsymbol{x} \leq \boldsymbol{z}$ has no solution.

Proposition 1.9 (Farkas lemma III).
Let $A \in \mathbb{R}^{m\times d}$, $\boldsymbol{z} \in \mathbb{R}^m$, $\boldsymbol{a}_0 \in (\mathbb{R}^d)^*$, and $z_0 \in \mathbb{R}$.

Then $\boldsymbol{a}_0\boldsymbol{x} \leq z_0$ is valid for all $\boldsymbol{x} \in \mathbb{R}^d$ with $A\boldsymbol{x} \leq \boldsymbol{z}$, if and only if
(i) *there exists a row vector $\boldsymbol{c} \geq \mathbb{O}$ such that $\boldsymbol{c}A = \boldsymbol{a}_0$ and $\boldsymbol{cz} \leq z_0$, or*
(ii) *there exists a row vector $\boldsymbol{c} \geq \mathbb{O}$ such that $\boldsymbol{c}A = \mathbb{O}$ and $\boldsymbol{cz} < 0$,*
or both.

Proof. The "if" part is easy to see: the existence of $\boldsymbol{x}$ with $A\boldsymbol{x} \le \boldsymbol{z}$ and $\boldsymbol{a}_0\boldsymbol{x} > z_0$ contradicts both (ii) (as in Farkas lemma I) and (i) (with a similar computation).

For the "only if" part, assume that neither (i) nor (ii) is satisfied. Then we conclude that there is no $\boldsymbol{b} \ge \mathbb{O}$ and $\beta \ge 0$ with $\boldsymbol{b}A = \beta\boldsymbol{a}_0$ and $\boldsymbol{b}\boldsymbol{z} < \beta z_0$: otherwise, (i) would be satisfied for $\boldsymbol{c} := \frac{1}{\beta}\boldsymbol{b}$ if $\beta > 0$, or (ii) for $\boldsymbol{c} := \boldsymbol{b}$, if $\beta = 0$.

Thus we can apply Farkas lemma I to compute

$$
\begin{array}{lll}
 & \nexists\,(\beta,\boldsymbol{b}) \ge (0,\mathbb{O}): & (\beta,\boldsymbol{b})\begin{pmatrix} -\boldsymbol{a}_0 \\ A \end{pmatrix} = \mathbb{O},\ (\beta,\boldsymbol{b})\begin{pmatrix} -z_0 \\ \boldsymbol{z} \end{pmatrix} < 0 \\
\overset{\text{FL I}}{\Longleftrightarrow} & \exists\,\boldsymbol{w} \in \mathbb{R}^d: & \begin{pmatrix} -\boldsymbol{a}_0 \\ A \end{pmatrix}\boldsymbol{w} \le \begin{pmatrix} -z_0 \\ \boldsymbol{z} \end{pmatrix} \\
\Longleftrightarrow & \exists\,\boldsymbol{w} \in \mathbb{R}^d: & A\boldsymbol{w} \le \boldsymbol{z},\ \boldsymbol{a}_0\boldsymbol{w} \ge z_0.
\end{array}
$$

Now we reformulate the condition that (i) does not hold, by introducing a slack variable γ, and then we apply Farkas lemma II to a problem in dual space:

$$
\begin{array}{llll}
\neg\text{(i)} & \Longleftrightarrow & \nexists\,(\gamma,\boldsymbol{c}) \ge (0,\mathbb{O}): & \gamma + \boldsymbol{c}\boldsymbol{z} = z_0,\ \boldsymbol{c}(-A) = -\boldsymbol{a}_0 \\
 & \Longleftrightarrow & \nexists\,(\gamma,\boldsymbol{c}) \ge (0,\mathbb{O}): & (\gamma,\boldsymbol{c})\begin{pmatrix} 1 & \mathbb{O} \\ \boldsymbol{z} & -A \end{pmatrix} = (z_0, -\boldsymbol{a}_0) \\
 & \overset{\text{FL II}}{\Longleftrightarrow} & \exists\begin{pmatrix} y_0 \\ \boldsymbol{y} \end{pmatrix} \in \mathbb{R}^{d+1}: & \begin{pmatrix} 1 & \mathbb{O} \\ \boldsymbol{z} & -A \end{pmatrix}\begin{pmatrix} y_0 \\ \boldsymbol{y} \end{pmatrix} \ge \begin{pmatrix} 0 \\ \mathbf{0} \end{pmatrix}, \\
 & & & (z_0, -\boldsymbol{a}_0)\begin{pmatrix} y_0 \\ \boldsymbol{y} \end{pmatrix} < 0 \\
 & \Longleftrightarrow & \exists\, y_0 \ge 0, \boldsymbol{y} \in \mathbb{R}^d: & A\boldsymbol{y} \le y_0\boldsymbol{z},\ \boldsymbol{a}_0\boldsymbol{y} > y_0 z_0.
\end{array}
$$

Now either we have $y_0 > 0$, then we put $\boldsymbol{x} := \frac{1}{y_0}\boldsymbol{y}$, and this satisfies $A\boldsymbol{x} \le \boldsymbol{z}$ and $\boldsymbol{a}_0\boldsymbol{x} > z_0$, or we have $y_0 = 0$, then we use the $\boldsymbol{w}$ constructed before (remember?) and put $\boldsymbol{x} := \boldsymbol{w} + \boldsymbol{y}$. This $\boldsymbol{x}$ satisfies $A\boldsymbol{x} = A\boldsymbol{w} + A\boldsymbol{y} \le \boldsymbol{z} + \mathbf{0} = \boldsymbol{z}$ and $\boldsymbol{a}_0\boldsymbol{x} = \boldsymbol{a}_0\boldsymbol{w} + \boldsymbol{a}_0\boldsymbol{y} > z_0 + 0 = z_0$. $\square$

The following, fourth and last version (but see the exercises) shows that the Farkas lemma can also be used to *separate* a point from a $\mathcal{V}$-polyhedron: if $\boldsymbol{x}$ is not contained in $P := \text{conv}(V) + \text{cone}(Y)$, then there is an inequality $\boldsymbol{a}\boldsymbol{x} \le \alpha$ satisfied by P, but not by $\boldsymbol{x}$.

Proposition 1.10 (Farkas lemma IV).
Let $V \in \mathbb{R}^{d\times n}$, $Y \in \mathbb{R}^{d\times n'}$, and $\boldsymbol{x} \in \mathbb{R}^d$.
Either *there exist $\boldsymbol{t}, \boldsymbol{u} \ge \mathbf{0}$ with $\mathbb{1}\,\boldsymbol{t} = 1$ and $\boldsymbol{x} = V\boldsymbol{t} + Y\boldsymbol{u}$,*
or *there exists a row vector $(\alpha, \boldsymbol{a}) \in (\mathbb{R}^{d+1})^*$ with $\boldsymbol{a}\boldsymbol{v}_i \le \alpha$ for all $i \le n$, $\boldsymbol{a}\boldsymbol{y}_j \le 0$ for all $j \le n'$, while $\boldsymbol{a}\boldsymbol{x} > \alpha$,*
but not both.

Proof. The "either" condition can be stated as

$$\exists \begin{pmatrix} \boldsymbol{t} \\ \boldsymbol{u} \end{pmatrix} \geq \begin{pmatrix} \boldsymbol{0} \\ \boldsymbol{0} \end{pmatrix} : \quad \begin{pmatrix} \mathbb{1} & \mathbb{O} \\ V & Y \end{pmatrix} \begin{pmatrix} \boldsymbol{t} \\ \boldsymbol{u} \end{pmatrix} = \begin{pmatrix} 1 \\ \boldsymbol{x} \end{pmatrix}$$

which by version II of the Farkas lemma is equivalent to

$$\begin{aligned} &\overset{\text{FL II}}{\iff} \quad \nexists\, (\alpha, -\boldsymbol{a}) \in (\mathbb{R}^{d+1})^* : && (\alpha, -\boldsymbol{a}) \begin{pmatrix} \mathbb{1} & \mathbb{O} \\ V & Y \end{pmatrix} \geq (\mathbb{O}, \mathbb{O}), \\ & && (\alpha, -\boldsymbol{a}) \begin{pmatrix} 1 \\ \boldsymbol{x} \end{pmatrix} < 0 \\ &\iff \quad \nexists\, (\alpha, -\boldsymbol{a}) \in (\mathbb{R}^{d+1})^* : && \alpha \mathbb{1} - \boldsymbol{a} V \geq \mathbb{O},\ \boldsymbol{a} Y \leq \mathbb{O},\ \boldsymbol{a}\boldsymbol{x} > \alpha, \end{aligned}$$

which is equivalent to the negation of the "or" condition. □

1.5 Recession Cone and Homogenization

Using the Farkas lemma, we can give an invariant description of some very important constructions (notably the recession cone and the homogenization of a convex set) and establish their basic properties. In Proposition 1.14 we will see that the homogenization $\mathrm{homog}(P)$ of a polyhedron coincides with the "associated cone" $C(P)$ that we used in Section 1.1.

Definition 1.11. Let $P \subseteq \mathbb{R}^d$ be a convex set. Then the *lineality space* of P is defined as

$$\mathrm{lineal}(P) \ := \ \{\boldsymbol{y} \in \mathbb{R}^d : \boldsymbol{x} + t\boldsymbol{y} \in P \text{ for all } \boldsymbol{x} \in P,\ t \in \mathbb{R}\},$$

and the *recession cone* (or *characteristic cone*) of P is defined as

$$\mathrm{rec}(P) \ := \ \{\boldsymbol{y} \in \mathbb{R}^d : \boldsymbol{x} + t\boldsymbol{y} \in P \text{ for all } \boldsymbol{x} \in P,\ t \geq 0\}.$$

Directly from the definition we can derive that $\mathrm{lineal}(P)$ is a linear subspace of $\mathbb{R}^d$. If we choose a complementary subspace U to $\mathrm{lineal}(P)$ (i.e., $U \cap \mathrm{lineal}(P) = \{\boldsymbol{0}\}$ and $U + \mathrm{lineal}(P) = \mathbb{R}^d$), then P can be decomposed as the Minkowski sum

$$P \quad = \quad \mathrm{lineal}(P) \ + \ (P \cap U)$$

of a linear subspace L and a convex set $P \cap U$ whose lineality space is zero: $\mathrm{lineal}(P \cap U) = \{\boldsymbol{0}\}$.

This reduction usually makes it possible to consider only polyhedra with lineality space $\{\boldsymbol{0}\}$, known as *pointed polyhedra* (if they are nonempty).

For $\mathcal{H}$-polyhedra we compute $\mathrm{lineal}(P(A, \boldsymbol{z})) = \{\boldsymbol{x} \in \mathbb{R}^d : A\boldsymbol{x} = \boldsymbol{0}\}$. So, except for a trivial linear summand we can usually consider polyhedra $P(A, \boldsymbol{z}) \subseteq \mathbb{R}^d$ for which A has full rank d.